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RESEARCH ARTICLE

# AI-Enabled Environmental Auditing and Sustainability Disclosure: Green Reporting Capability as Mediator and Digital Compliance Culture as Moderator

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## ABSTRACT

Small and medium-sized enterprises (SMEs) represent a vital yet vulnerable segment of emerging economies, requiring strategic resources and dynamic capabilities to achieve long-term sustainability. Drawing on resource-based theory (RBT) and the Legitimacy Theory, this study examines the impact of AI-enabled environmental auditing (AIEA) on corporate sustainability disclosure (CSD), investigating the mediating role of green reporting capability (GRC) and the moderating role of digital compliance culture (DCC). Using structural equation modelling (SEM) on survey data from 344 SMEs operating in manufacturing and service sectors in South Asia (Pakistan, India, and Bangladesh), the findings reveal that AIEA significantly enhances both GRC and CSD, as well as the relationship between AIEA and CSD, highlighting the strategic role of a strong DCC. The results further indicate that GRC partially mediates the relationship between AIEA and CSD, underscoring the importance of robust green reporting systems in translating AI-based auditing practices into improved sustainability transparency. The study contributes to the sustainability and digital governance literature by demonstrating how the synergy between AI-enabled auditing mechanisms and DCC fosters credible, accountable, and sustainability-oriented disclosure practices among SMEs in emerging economies.

## 1 | Introduction

The increased demands of global sustainability issues such as climate change and environmental destruction through unsustainable industrial development have increased the significance of reporting environmental issues and good corporate governance practices in the global arena. The Small and medium-sized enterprises (SMEs) have been one of the factors contributing to economic growth in the South Asian emerging economies, but they are very susceptible to environmental pressures, institutional weaknesses, and compliance-related constraints (Quan 2025). In spite of their economic value, SMEs have a high tendency to be severely constrained (limited financial resources, poor governance mechanisms, lack of digital maturity, etc.) and fail to update to sustainable business practices (He et al. 2025).

In this regard, AI-enabled environmental auditing (AIEA) has become a disruptive technological process that can improve environmental monitoring, performance analysis, and sustainability reporting (Gebhardt et al. 2023). AIEA helps to support real-time analytics, check environmental data automatically, predict risks and enhance corporate practices in line with global sustainability agendas (Aldousari 2025; Zhou and Bu 2025). Although in the developed world, large companies are adopting AI-based sustainability solutions more and more, South Asian SMEs are still falling behind because

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of resource limitations, low adaptive potential, and high institutional informality (Naveed et al. 2025). As a result, the sustainability information reporting among SMEs is rather manual and highly inconsistent, which further increases information asymmetries and lack of trust in stakeholders (Wei et al. 2025).

This paper examines how AI-enabled Sustainability Auditing can be utilized to improve corporate sustainability disclosure (CSD), which is one of the most significant processes by which companies report green, social, and governance (ESG) promises. CSD is crucial so as to enhance transparency, organisational legitimacy, and a sense of trust between the organisation and its regulators, investors and the entire society (Lu and Li 2025). Based on the resource-based theory (RBT), the AI-Enabled Enviromental Auditing is understood as the strategic technological resource that can be used to create sustainable competitive advantage. Likewise, the Legitimacy Theory describes the role of improved disclosure practices in strengthening the social acceptability of organisations. This paper reflects on the technological and organisational drivers of sustainability disclosure in SMEs on a combination of theoretically based and contextual variables, which provides a distinct perspective in uniting technology adoption and ESG performance.

This study is innovative in the fact that it does not simply present AI-Enabled Enviromental Auditing as a digital tool, but as an enabler of credible sustainability reporting. Although there are earlier researches in general environmental auditing or CSR practices, not many have empirically related AI- enabled systems to the quality, reliability and transparency of sustainability disclosures (Almaqtari 2024). The research also provides the concepts of green reporting capability (GRC) as the mediating variable because the use of advanced auditing technologies is not sufficient to ensure credible disclosure without substantial internal systems to process and communicate environmental information (Gazi et al. 2024; Lai 2025). Also, digital compliance culture (DCC) is explored as a moderating factor, which is based on ethical orientation, maturity in digital governance, and dedication to compliance-driven transparency, which in turn improve the effectiveness of AI auditing (Tomar 2025; Tarawiru et al. 2025). The application of these variables is exacerbated by the South Asian SME environment because of the disparate regulatory application and the lack of a well-established institutional structure (Li and Jin 2024). The growing demands of world consumers and foreign investors force SMEs in Pakistan, India, and Bangladesh to enhance the quality of sustainability reporting (Aldhi et al. 2025). Notwithstanding these pressures, the majority of SMEs still use manual environmental audits that restrict the reliability, timeliness, and comparability of their disclosure of sustainability. The literature available is largely related to CSR, environmental reporting, or broadly speaking, digital transformation in large institutions, whereas there is limited empirical evidence regarding the application of AI-based environmental auditing in the context of SMEs (Tarawiru et al. 2025; Camilleri 2022). Digital transformation has been considered extensively; however, the study of the particular ways in which AI-based technologies can boost the believability and disclosure of sustainability has received little focus (Khan et al. 2024; Kant et al. 2025). Besides, little research on the disclosure of sustainability focuses on the technological processes behind the scene, including AI-charged auditory systems, which enhance the precision, verification, and dependability of environmental information (Saleh et al. 2025).

Although the limitation in SME resources is well reported (Shen et al. 2025), earlier studies have mostly ignored the role of AI-enabled auditing in mitigating the same in developing economies, which have poor institutional and infrastructural environments (Chen and Zhang 2025; Elshaer et al. 2025; Landi et al. 2022). Notably, there are no combined empirical frameworks that analysed the mediating or moderating effect of GRC or DCC between AI-Enabled Enviromental Auditing and sustainability disclosure (Sivaiah and Vinodan 2025; Damnjanović et al. 2025). Furthermore, the current sustainability disclosure systems are also mainly Western-oriented and do not correspond to the South Asian market realities, in which SMEs are already operating under the extreme resource constraints and starting to be held to higher accountability standards (de Almeida and Oliveira 2024; He et al. 2025).

To address these gaps, this study proposes the following objectives:

1. To find out how AIEA affects the CSD of SMEs in South Asia.
2. To determine the role of GRC in mediating between the AIEA and CSD relationship.
3. To test the moderating effect of DCC in the relationship between AIEA and CSD.

The study has strong theoretical and practical value. In theory, it contributes to the literature on digital transformation and environmental management, as it frames AI as a strategic resource that can increase firm legitimacy by promoting better transparency. In practice, it offers actionable policy implications that are practical on how South Asia SMEs can enhance environmental responsibility with cost-effective and smart auditing tools. This paper is unique in that it integrates AI auditing technologies, green reporting functionalities, and DCC into an integrated framework empirically examined across multiple South Asian economies. Compared to earlier literature, which investigates technological or cultural factors independently, this study reveals the synergistic effects of technological and cultural factors on sustainability transparency, providing a more comprehensive understanding of digital sustainability transformation. Using a quantitative research design and SEM, the article examines survey data collected from 344 SMEs operating in environmentally sensitive industries in Pakistan, India, and Bangladesh. To ensure empirical rigor, appropriate procedures were employed to ensure construct validity and reliability. The research advances the literature by: (1) proposing AIEA as a new technological capability in the sustainability disclosure study; (2) underlining the mediation of GRC in changing technological inputs into sustainability outcomes; and (3) demonstrating the significance of DCC as a situational requirement for successful AI-based environmental management. Collectively, these inputs produce the richness of the discourse of sustainable digital transformation and offer SMEs and policymakers strategic avenues that can foster ethical, transparent, and environmentally responsible business approaches.



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## 2 | Theoretical Foundations and Hypothesis Development

The integration of RBT, Legitimacy theory and dynamic capabilities theory (DCT) provides a multi-level theoretical lens for understanding the relationship between AIEA and CSD, with GRC as a mediator and DCC as a moderator. All theories present a unique yet complementary explanation that all help to promote the conceptual originality of the model. According to RBT, sustained firm performance stems from the effective deployment of valuable, rare, inimitable and non-subtitutable resources (Chae et al. 2014). AIEA meets these criteria by integrating digital infrastructure, predictive algorithms, real-time enviromental data and expert knowledge systems to enhance the accuracy and reliability of environmental assessments (Fosso-Wamba et al. 2024). These capabilities based on AI decrease the information asymmetry, enhance environmental surveillance, and enhance the efficiency of compliance in SMEs (Gazi et al. 2024). However, as explained by RBT emphasizes that resource possession alone does not guarantee superior outcomes unless firms develop complementary capabilities to transform resources into strategic value (Quan 2025). In this context, GRC represents a complementary organizational capability that enables SMEs to process, structure and translate AIEA-generated insights into credible sustainability disclosures (Tarawiru et al. 2025). GRC transforms AI-based intelligence into the decision-ready and externally validated reports through the use of routinized processes, professionalism, and standardized documentation. Therefore, RBT clarifies the process through which AIEA becomes a strategic resource the influence of which on CSD becomes a tangible reality only with the help of developed reporting facilities. Legitimacy theory further explains sustainability disclosure as a response to social expectations, institutional pressures and the pursuit of organizational acceptance (O'donovan 2002). The demand on SMEs to be more transparent is growing in emerging economies by regulators, customers, and world supply chains (Gazi et al. 2024). AIEA enhances the legitimacy-seeking behaviour of firms by generating auditable, objective and credible evidence of environmental performance which can be communicated to the stakeholders (Aldhi et al. 2025). However, credible legitimacy outcomes require more than data Availability; enviromental intelligence must be structured into stakeholder-oriented disclosures that minimize ambiguity. In this regard, GRC functions as an organizing mechanism that transforms AI-generated enviromental insights into coherent, stakeholder-centred sustainability reports (Damnjanović et al. 2025). DCC also strengthens the motives of legitimacy by turning the ethical digital practices, transparency and digital responsibility into institutionalised ones (Li and Jin 2024). When employees internalize digital compliance norms and reporting protocols, the accuracy and credibility of sustainability disclosures improve, thereby strengthening organizational legitimacy. In this way, the Legitimacy Theory explains the increase in CSD caused by AIEA and the role of DCC in enhancing the motivation. DCT provides an evolutionary perspective, explaining how firms sense environmental changes, seize emerging opportunities and reconfigure organizational resources overtime (Teece 2018). AIEA augments sensing capabilities of SMEs by allowing them to constantly monitor the threats to the environment, infractions of regulations, and inefficiencies of sustainability (Neiroukh and Çağlar 2025). This real-time sensing will enable firms to predict institutional shifts and seize opportunities in the context of sustainability (Kant et al. 2025). GRC is a seizing capacity since it helps organisations to combine, assess and formalize outputs of AIEA into structured disclosures that can aid strategic decision-making (Tomar 2025). Over time, routine sustainability reporting evolves into a reconfiguring capability, enabling firms to redesign internal processes, revise reporting routines and Insitituationalize transparency as an ongoing organizational competence (Chen and Zhang 2025). DCC enhances these dynamic capabilities by enhancing digital preparedness, compliance focus, and technological flexibility (Aldousari 2025). A vigorous DCC fosters faster AIEA implementation, deeper digitization of the reporting processes, and facilitates constant reconfiguration as regulatory and environmental demands are changing. The integration of RBT, Legitimacy Theory and DCT offers a theoretical explanation that is multi-dimensional and that makes the model more novel and has a greater explanatory potential. RBT considers AIEA as one of the strategic resources and GRC as value-realizing capability. Legitimacy Theory explains the institutional and social reasons behind how AIEA outputs should become transparent sustainability disclosures and how DCC increases such legitimacy-based behaviours. DC provides a more process-focused dimension of additional explanation of how AIEA is associated with learning, sensing, capture of opportunities, and the reconfiguration of organisations, which in the long run enhances the sustainable reporting quality. In these frameworks, DCC can be discussed as an influential contextual modulator which enhances the resources implementation, improves the disclosure-driven by legitimacy, and promotes the development of capabilities. Together, the CSD is influenced by technological resources, the formation of capabilities, institutional pressures, and digital preparedness that the organisational culture carries.

### 2.1 | AI-Enabled Environmental Auditing

AIEA refers to the application of machine learning, sensor analytics, natural language processing and automated validation systems to enhance environmental assessment and compliance monitoring (Shen and Badulescu 2025; Gazi et al. 2024). AIEA processes large-scale enviromental datasets with greater precision and reliability and responds to real-time data variations more effectively than manual auditing approaches (Almaqtari 2024). Through automated pattern recognition and anomaly detection, AIEA rapidly identities enviromental risks, operational inefficiencies and compliance breaches, thereby reducing measurement errors and strengthening evidence-based sustainability evaluations (Shen et al. 2025). Moreover, AIEA generates transparent and traceable data streams, which enhance the credibility and verifiability of sustainability reporting (Tomar 2025). AIEA further strengthens internal controls and supports predictive environmental risk management by reducing human bias and enabling continuous and repeated compliance checks (Naveed et al. 2025; Bindeeba et al. 2025). These features position AIEA as a transfomative tool for enhancing corporate transparency and sustainability performance (Lai 2025). By enhancing firm's capacity to disclose timely, credible and data-driven, AIEA strengthens organizational alignment with the expectations of regulatory expectations and broader social accountability requirements (Quan 2025). Overall, AIEA constitutes a strategic



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technological capability that facilitates green reporting processes and enhances sustainable accountability within contemporary organizations (Khan et al. 2024).

### 2.2 | Green Reporting Capability

GRC refers to an organisation's ability to systematically collect, process, structure, and disseminate environmental information to support managerial decision-making (Quan 2025). It integrates technical, procedural and human capabilities that enable firms to transform raw environmental data into reliable and stakeholder-oriented sustainability reports (Aldhi et al. 2025). GRC strengthens sustainability governance by embedding environmental indicators into managerial decision-making processes and supporting systematic data management and reporting practices (He et al. 2025). GRC enhances the consistency, comparability and credibility of sustainability disclosures data through effective documentation, digital integration and inter-functional coordination (Chen and Zhang 2025). Empirical evidence suggests that robust reporting capabilities not only improve disclosure quality but also support compliance with regulatory and global sustainability standards (Sivaiah and Vinodan 2025). Furthermore, GRC promotes organisational learning by employees to interpret environmental trends, identify performance gaps and propose mitigation strategies for long-term ecological improvement (Chen and Zhang 2025). The literature increasingly conceptualizes GRC as a strategic capability that links operational processes with broader sustainability objectives by converting environmental intelligence into structured and transparent reporting outputs (He et al. 2025). In highly dynamic environments, effective Green reporting has become essential for meeting stakeholder expectations and demonstrating responsible environmental stewardship (Khan et al. 2024). Accordingly, GRC represents a critical organisational capability that institutionalises sustainability reporting, strengthens environmental accountability and enhances competitiveness in data-driven, and sustainability-oriented global markets (Neiroukh and Çağlar 2025).

### 2.3 | Digital Compliance Culture

DCC refers to a set of shared organisational values, expectations, and norms that promote the ethical and responsible use of digital technologies to support regulatory compliance and transparency (Shen et al. 2025). It reflects a cultural environment where employees consistently utilize digital tools, prioritise data, and adhere to environmental and safety regulations (Almaqtari 2024). A strong DCC aligns digital protocols with sustainability commitments, compliance responsibilities, and guiding employees towards consistent reporting practices and maintaining organisational integrity (He et al. 2025). By mitigating resistance to new technologies, DCC fosters digital readiness, continuous learning, and leadership-driven compliance practices (Shen and Badulescu 2025). Organizations with a good DCC achieve higher technology adoption, enhanced data management, and quicker responsiveness to regulatory changes (Chen and Zhang 2025). Moreover, DCC improves interdepartmental cooperation by establishing standardised digital guidelines that streamline information flow and align compliance expectations (Li and Jin 2024). Studies indicate that a strong digital culture determines the effectiveness of technology investments, including AIEA, in improving organisational performance and disclosure quality (Tomar 2025). Institutionalising digital compliance norms enhances organisational transparency, strengthens sustainability practices, and increases environmental accountability (Kant et al. 2025). Consequently, DCC represents a critical cultural capacity that facilitates digital transformation, reinforces organizational legitimacy, and promotes structured governance in regulated digital environments (Aldhi et al. 2025).

### 2.4 | Corporate Sustainability Disclosure

CSD refers to the systematic reporting of a firm's ESG performance to its stakeholders (Naveed et al. 2025). CSD enables organisations to communicate resource use, environmental impacts, and compliance initiatives, thereby reducing information asymmetry and supporting informed decision-making (Shen et al. 2025). CSD promotes accountability, strengthens ethical responsibilties and ensures alignment with global sustainability frameworks including GRI, SASB, and integrated reporting standards (Aldhi et al. 2025). Well-developed CSD systems enhance stakeholders trust, improve corporate reputation and signal long-term commitment to environmental responsibility (Lai 2025). High-quality disclosures foster investor confidence, support regulatory compliance and strengthen performance within global supply chains (Naveed et al. 2025). Internally, CSD informs strategic planning by providing managers with critical insights for risk management and resource allocation (Tomar 2025). Contemporary research recognize sustainability reporting not as a voluntary activity, but as a strategic tool for legitimacy formation, competitive differentiation, and regulatory compliance in environmentally conscious markets (Quan 2025). Effective Long-term CSD ensures reliability, timeliness, and transparency by combining accurate environmental information with robust organisational capabilities and a supportive cultural environment (Zhou and Bu 2025). Ultimately, CSD is a core organisational process through which firms communicate sustainability performance, and demonstrate compliance and legitimacy within highly monitored institutional contexts (Aldhi et al. 2025).

### 2.5 | AIEA on Corporate Sustainability Disclosure

AIEA encompassing AI, machine learning and automated environmental assessment tools, enhances CSD by generating accurate, real-time and verifiable enviromental data (Gazi et al. 2024; Damnjanović et al. 2025). From a RBT perspective, AIEA represents a valuable, rare and inimitable capability that enhances a firm's ability to detect environmental indicators, identify compliance gaps, and assess operational efficiency, thereby improving the accuracy and reliability of disclosures (Lai 2025). Transforming AI-generated insights into standardised, audit-ready reporting formats, allows firms to meet heightened requirements for accuracy and comparability in sustainability disclosures. According to Legitimacy Theory, AIEA enables organisations to provide timely and credible evidence of environmental compliance, demonstrating their commitment to sustainabile practices. This reduces information asymmetry and



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increases stakeholder trust by aligning organizational reporting with regulatory and societal expectations (Kant et al. 2025; Tomar 2025; Wei et al. 2025). DCT highlights how AIEA enhances a firm's capacity to sense environmental risks, integrate cross-functional information, and continuously refine reporting patterns. AI-based learning processes allow organisations to adapt disclosure content in response to evolving regulatory requirements and stakeholder expectations, thereby enhancing both flexibility and relevance (Sivaiah and Vinodan 2025; He et al. 2025). Together, these theoretical perspectives indicate that AIEA functions not merely as a technological tool but a strategic facilitator of transparent, reliable, and high-quality sustainability disclosures. AIEA directly enhances the quality of CSD by strengthening environmental accountability, reducing reporting errors, and promoting organisational legitimacy (Damnjanović et al. 2025).

**Hypothesis 1.** *AIEA is positively and directly associated with CSD.*

## 2.6 | Mediating Role of Green Reporting Capability

GRC refers to a firm's ability to collect, process, and present environmental data in a transparent, reliable, and decision-orientwd manner (He et al. 2025). GRC transforms raw and unverifiable environmental information into credible and decision-supportive reports that enhance transparency, accountability, and managerial decision-making. From a RBT perspective, GRC represents a valuable, firm-specific and difficult-to-imitate capability that transforms internal environmental knowledge into high-quality disclosures, thereby generating competitive and legitimacy-enhancing advantages (Li and Jin 2024). GRC employs standardised reporting routines, internal verification mechanisms and cross-functional coordination to ensure that environmental performance measures are accurate, consistent and aligned with requirements and stakeholder expectations (Shen et al. 2025). According to Legitimacy Theory, GRC enhances the organisational legitimacy by providing credible evidence of regulatory compliance and proactive environmental stewardship, thereby strengthening stakeholder trust and corporate reputation (Gazi et al. 2024; Aldhi et al. 2025). GRC reduces information asymmetry, mitigates reputational risk, and signals organisational commitment to sustainability goals by translating complex environmental data into comprehensible and verifiable reports (Lu and Li 2025). DCT further explains how GRC enables firms to integrate environmental expertise into reporting systems, identify improvement opportunities, and reconfigure internal practices to address evolving sustainability demands (He et al. 2025; Chen and Zhang 2025). This iterative capability allows organisations to continuously revise reporting templates, incorporate additional performance metrics and align disclosures with both current performance and future sustainability objectives. The integration of RBT, Legitimacy Theory, and DCT provides a multi-level explanation of GRC's influence on CSD, capturing resource value creation, insititutional legitimacy and continuous capability development. Accordingly. GRC functions as a strategic organizational capability that enhances the quality, reliability, and completeness of CSDs, thereby justifying the following hypothesis:

**Hypothesis 2.** *GRC is positively and directly associated with CSD.*

The mere adoption of AIEA in contemporary organisations does not automatically guarantee credible and high quality CSD. The effective implementation of AIEA depends on a firm's ability to interpret and standardise communicate environmental data, a role fulfilled by GRC (Lai 2025). RBT posits that while AIEA consitutes a valuable and rare technological asset, it's potential to enhance CSD materializes only when organisations posses the capability to codify and operationalise environmental information (Elshaer et al. 2025). GRC acts as an intermediate capability that transforms AI-generated audit reports into structure, trustworthy, and decision-useful sustainability reports (Shen et al. 2025). In the absence of such reporting capabilities, even advanced AI solutions may produce incomplete or unreliable, information that fails to gain stakeholder confidence (Tarawiru et al. 2025). From a Legitimacy Theory perspective, stakeholders evaluation of corporate transparency depends not only on data availability but also on its interpretability, verifiability, and alignment with societal norms (Quan 2025). GRC ensures that the insights generated through AIEA are communicated using credible and standardised reporting formats, thereby enhancing stakeholder confidence and institutional legitimacy (Fosso-Wamba et al. 2024; Neiroukh and Çağlar 2025). DCT further clarifies the mediating role of GRC: while AIEA strengthens a firm's sensing capability by identifying environmental risks and performance gaps, GRC enables the organization to seize and reconfigure these insights into actionable reporting processes (Kant et al. 2025). Through iterative learning, cross-functional collaboration, and continuous optimization, GRC facilitates the transformation of AI-generated environmental data into holistic, timely, and decision-relevant sustainability disclosures (Damnjanović et al. 2025). Collectively, these theoretical perspectives demonstrate that the impact of AIEA on CSD is largely indirect and becomes substantially stronger when organisations emply robust internal reporting capabilities to organise, contextualise, and communicate environmental information. According, GRC serves as a critical mediating mechanism that links AI-enabled autditing resources with tangible sustainability disclosure outcomes (Almaqtari 2024).

**Hypothesis 3.** *GRC mediates the relationship between AIEA and CSD.*

## 2.7 | Moderating Role of Digital Compliance Culture

DCC describes the organisation values, norms and behaviours that support ethical, accountable and technology-enabled compliance practices in sustainability reporting (Gazi et al. 2024). In firms with a strong DCC, digital tools, compliance systems, and sustainability standards are integrated across organizationsal levels, ensuring that employees consistently adhere to reporting and regulatory requirements (He et al. 2025). From a RBT perspective, DCC can be conceptualized as a socially embedded capability that enhances the effective utilization of technological and informational resources such as AI-enabled auditing tools (Li and Jin 2024). The institutionalization of digital compliance practices ensures the systematic generation



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of accurate, complete and well-documented sustainability data, thereby directly enhancing the quality, credibility and reliability of CSD (Elshaer et al. 2025). Legitimacy Theory further explains that DCC strengthens organizational credibility by aligning disclosure practices with stakeholder expectations (Quan 2025). A strong DCC signals effective governance and enviornmental responsibility, thereby assuring regulators, investors, and the public that sustainability is addressed in a transparent and accountable manner (Tomar 2025). It minimizes chances of misreporting, selective disclosure or greenwashing and, as a result, stakeholders trust sustainability reports more (Shen and Badulescu 2025; Lai 2025). DCT positions DCC as a facilitator of organisational agility in sustainability reporting. By fostering digital literacy, process standardization and proactive oversight, DCC enables firms to respond to evolving compliance requirements, enhance disclosure quality, and continuously reconfigure reporting practices (Wei et al. 2025; Aldhi et al. 2025). Taken together, these theoretical perspectives indicate that DCC is not merely a favorable cultural attribute but a strategic organisational capabilitythat supports the effective deployment of technological resources (RBT), enhances stakeholder legitimacy (Legitimacy Theory) and promotes adaptability and flexibility in sustainability reporting processes (DCT).

**Hypothesis 4.** *DCC is positively associated with CSD.*

In contemporary organisations, the integration of advanced technologies, including AIEA, does not automatically result in high-quality CSD. The effective transfomation of AI-generated insights into meaningful CSD largely depends on the organisational environment, particularly DCC, which encompasses values, norms, and behaviours that support ethical, accountable, and technolgy-driven compliance (Khan et al. 2024). Firms with a strong DCC ensure that AI-generated environmental data are regularly verified, interpreted, and systematically incorporated into formal reporting procedures (Elshaer et al. 2025). Such cultural internalization facilitates the acceptance and effective use of digital tools, enabling organizations to convert technological outputs into credible, timely and actionable sustainability disclosures. From a RBT perspective, DCC complements AI-related resources by enabling their effective deployment and value realization. Even highly sophisticated AI systems may fail to deliver value in the absence of a supportive digital culture, as they may remain underutilized, mistrusted, and improperly applied. Conversely, in organisations characterized by high DCC, AI-based insights are effectively translated into operational practices, resulting in more accurate, comprehensive, and trustworthy sustainability disclosures (Gazi et al. 2024; Naveed et al. 2025). Legitimacy Theory suggests that DCC enhances stakeholder trust by signalling proactive governance and environmental responsibility. Once the results of AI auditing are intertwined with the culture of digital compliance, this sends a message to regulators, investors and society that sustainability reporting is technologically sound and ethically regulated (Shen and Badulescu 2025). This dual signalling strengthens organisational legitmacy and stakeholder confidence, thereby reinforcing transparency and accountability in sustainability reporting. DCT further emphasises that DCC enhances organizational learning, flexibility and cross-functional integration. By institutionalizing digital compliance standards, firms can continuously adapt reporting practices, anticipate emerging regulatory requirements, and utilise AI-driven insights to improve the quality and relevance of sustainability reporting (He et al. 2025; Damnjanović et al. 2025).

**Hypothesis 5.** *DCC positively moderates the relationship between AIEA and CSD, such that the effect of AIEA on CSD is stronger when DCC is high.*

Similarly, the mere presence of GRC does not automatically ensure the comversion of environmental information into quality sustainability reports. Organisational culture-particularly norms and values that support digital compliance, largely determines the effectiveness of GRC (Tarawiru et al. 2025). DCC creates enabling conditions under which GRC can be optimally operationalised, promoting the consistent verification, standardisation, and timely sustainability reporting (Chen and Zhang 2025; Quan 2025). From a RBT perspective, DCC complements GRC by enhancing the organization's ability to transform technical capabilities into meaningful and adding disclosure outputs (Fosso-Wamba et al. 2024). Even well-developed reporting procedures may fail in the absence of a supportive culture that encourages employees to adopt data systematically and maintain transparency. A high level of DCC ensures that GRC outputs are credible, comprehensive and practically applicable (He et al. 2025). Legitimacy Theory further explains that stakeholders evaluate sustainability disclosures not only based on the existence of reporting capabilities but also on their fair, consistent and transparent implementation (Tomar 2025). A strong DCC encourages sound reporting practices, thereby strengthening stakeholder confidence andorganisational legitimacy (Shen and Badulescu 2025). From a DCT perspective, DCC facilitates continuous improvement and adaptive learning in sustainability reporting. Organizations can formalize feedback mechanisms, refine reporting templates, and align disclosure practices with evolving stakeholder and regulatory demands, thereby maximizing the effectiveness of GRC in producing high-quality sustainability reports (Aldhi et al. 2025; Gazi et al. 2024).

**Hypothesis 6.** *DCC positively moderates the relationship between GRC and CSD, such that the effect of GRC on CSD is stronger when DCC is high.*

Building on the above hypotheses, the conceptual framework is developed as shown in Figure 1.

## 3 | Methods

### 3.1 | Operationalization, Sample, and Data Collection

Measurement scales of the critical constructs employed in this research study were derived on the basis of the vast literature review and pre-tested so as to make them contextually relevant and intelligible to SMEs in South Asia. The items were modified into the already tested studies, and slight adjustments were made to the unique operational, environmental, and regulatory circumstances of the SMEs in Pakistan, India, and Bangladesh. A five-point Likert scale with values of 1 = strongly disagree to



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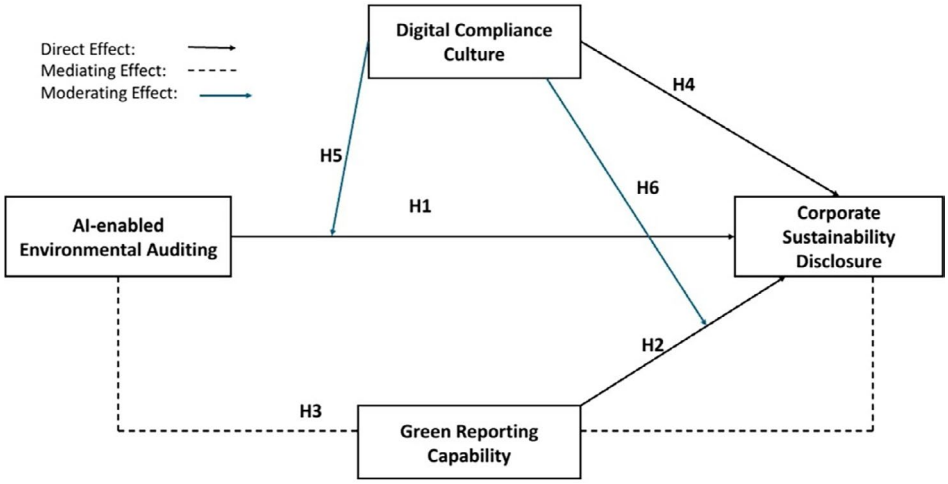



**FIGURE 1** | Conceptual framework.

5 = strongly agree was utilized to measure all constructs, enabling the respondents to respond to how much they agree with a particular statement.

### 3.2 | Recruitment and Sample Characteristics of the Participants

The population targeted was SMEs in Pakistan, India, and Bangladesh. Stratified purposive sampling method was used to obtain sufficient coverage in the three leading South Asian economies as well as various industries (manufacturing, service, and retail). The sampling frame was the official SME directories, which comprised the Small and Medium Enterprise Development Authority (SMEDA) Pakistan, Bangladesh Investment Development Authority (BIDA), and India SME Yellow Pages, which contain a complete list of registered SMEs.

Based on these databases, 1825 SMEs were selected and approached by email and LinkedIn in April to September 2025 with the use of a self-administered questionnaire. The amenity of official directories also improved the credibility of the sampling frame whereas the stratification based on the countries made sure that the proportional regional representation existed in South Asia. Following the background checking on completeness and response validity, 312 useful responses were made, which gave an effective response rate of 17.3. The last sample is the representation of the South Asian SME segment, which is made up of 132 Pakistani SMEs (42.3%), 108 Indian SMEs (34.6%), and 72 Bangladeshi SMEs (23.1%). Such distribution reflects economic and industrial diversity of the region and enhances the possibility to generalize the findings of cross-national quality. Key informants were chosen as owners, CEOs, directors and top managers due to their strategic knowledge, environmental and technological knowledge as these would offer correct information on organisational practises oriented on AIEA, reporting capabilities, compliance culture and sustainability disclosure. As it can be observed in Table 1, the sample is diverse in the demographic and the organisational terms, which increases the representativeness and strength of the empirical analysis.

### 3.3 | Measurement Model

To achieve reliability and validity, all constructs were operationalized into several items that were based on previous studies that were validated. AIEA was calculated on three levels: AI integration, environmental monitoring, and environmental risk assessment. Two items were used to measure each of the dimensions, yielding a six-item scale. The adapted items were used in Wei et al. (2025) and Tomar (2025), which are related to the digital auditing processes and the environmental management practices. In the past, it was found that this construct had high internal consistency (Kant et al. 2025; $r = 0.86$). DCC was measured in a seven-item scale created by Fosso-Wamba et al. (2024) and Chen and Zhang (2025) comprising organisational digital ethics, compliance norms, and technological integrity. The scale identifies the degree to which the firms integrate ethical, accountable, and technology-driven behaviours in their culture. GRC, with a four-item scale according to Aldousari (2025) and Lai (2025), is the capacity of the firm as regards to its capability to gather, analyse, and report environmental information in a systematic manner. This construct is the ability to transform the raw environmental data into orderly, actionable sustainability information. CSD was measured with four items based on the synthesis of Tomar (2025) and Li and Jin (2024) of the level, quality, and transparency of sustainability reporting according to the GRI standards. The reliability and validity of all scales were verified, and the information about the survey items and the statistics were provided in Table 2. Using the measures that have been tested and validated and recruiting knowledgeable respondents by using official SME directories, the study ensured that both constructs and data were relevant contextually, accurate, and representative of the South Asian SME industry.

### 3.4 | Quality Check

After checking that the sample complied with adequacy criteria, the quality of the measurement was first evaluated with the help of Exploratory Factor Analysis (EFA). The Kaiser Meyer Olkin



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**TABLE 1** | Profile of the respondents and firms.

| Categories | Frequency | Percentage |
|---|---|---|
| Gender | | |
| Male | 239 | 76.6 |
| Female | 67 | 21.5 |
| Prefer not to say | 6 | 1.9 |
| Age | | |
| 18–24 | 22 | 7.1 |
| 25–34 | 91 | 29.2 |
| 35–44 | 106 | 34.0 |
| 45–54 | 63 | 20.2 |
| 55+ | 30 | 9.5 |
| Education | | |
| High school | 25 | 8.0 |
| College/Diploma | 88 | 28.2 |
| Undergraduate degree | 107 | 34.3 |
| Master's degree | 74 | 23.7 |
| Doctorate/Professional | 18 | 5.8 |
| Position | | |
| Owner/CEO/Proprietor | 138 | 44.2 |
| Director | 68 | 21.8 |
| Manager | 90 | 28.8 |
| Supervisor | 16 | 5.2 |
| Annual Turnover (USD) | | |
| Under $1 million | 205 | 65.7 |
| $1 million–$5 million | 63 | 20.2 |
| $5 million–$10 million | 31 | 9.9 |
| Over $10 million | 13 | 4.2 |
| Number of employees | | |
| Under 10 | 182 | 58.3 |
| 10–49 | 82 | 26.3 |
| 50–249 | 48 | 15.4 |
| Industry type | | |
| Retail | 154 | 49.4 |
| Manufacturing | 68 | 21.8 |
| Service | 90 | 28.8 |
| Firm age | | |
| Under 3 years | 71 | 22.8 |
| 3–5 years | 75 | 24.0 |
| 6–10 years | 67 | 21.5 |
| 11–15 years | 49 | 15.7 |
| Over 15 years | 50 | 16.0 |
| Country representation | | |
| Pakistan | 132 | 42.3 |
| India | 108 | 34.6 |
| Bangladesh | 72 | 23.1 |
| Total | 312 | 100% |

(KMO) measure of sampling adequacy value was above the recommended value of 0.70, which indicated that the data was appropriate to undergo factor analysis. The Test of Sphericity by Bartlett was statistically significant ($p < 0.001$), which in turn supported factorability. The EFA generated a complete and theoretically consistent factor structure using ML extraction with Promax rotation on all the 21 measured items. The loading of all factors was more than 0.50, and the cross-loadings were low and insignificant, which indicates a good item-to-construct concordance and unidimensionality in the exploratory stage. After EFA, the measurement model was strictly verified on the basis of Confirmatory Factor Analysis (CFA) in the framework of SEM. Construct validation only mentions standardized loadings of SEM. All the standardized factor loadings were more than the recommended amount of 0.50, which validates high indicator reliability. The convergent validity was determined by the fact that the values of the Average Variance Extracted (AVE) were greater than 0.50 (He et al. 2025; Almaqtari 2024; Hair et al. 2019). Composite Reliability (CR) was used to measure construct reliability, and all CRs were over the recommended value of 0.70, which means high internal consistency (Hair et al. 2019). The discrimination validity was assessed in two steps. To start with, pairwise correlations between constructs had values ranging between 0.218 and 0.652 (Table 3), which are significantly lower than the conservative value of 0.85, which shows that constructs are sufficiently differentiated (Kline and Tamer 2016). Second, the Fornell-Larcker criterion was used by means of comparing squared inter-construct correlations with the AVE values (Table 4). In each of these instances, the squared correlations were below the AVE values of the concerned constructs, which indicates the existence of clear discriminant validity (Kant et al. 2025). In general, the measurement model exhibits excellent psychometric quality, such as adequate sampling adequacy, factor structure is clear at the exploratory phase, standardized factors loading in SEM is substantial, convergent validity is acceptable, composite reliability is strong, and there is already established discriminant validity. These findings establish the fact that the measurement model satisfies the fundamental needs of reliability and validity to further structural analysis.

### 3.5 | Normality Check

Normality testing is essential to test the distribution characteristics of the data obtained. Univariate and multivariate normality



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**TABLE 2** | Measurement scale and reliability statistics.

| Construct, items and code, source and reliability statistics | Loading (EFAa) | Loading (CFAb) |
|---|---|---|
| AI-enabled environmental auditing (AIEA): Cronbach's Alphac = 0.93; AVEd = 0.701; CRe = 0.921 | | |
| AIEA1 | 0.78 | 0.82 |
| AIEA2 | 0.83 | 0.85 |
| AIEA3 | 0.90 | 0.91 |
| AIEA4 | 0.85 | 0.88 |
| AIEA5 | 0.79 | 0.82 |
| AIEA6 | 0.87 | 0.89 |
| Digital compliance culture (DCC): Cronbach's alpha = 0.94; AVE = 0.687; CR = 0.936 | | |
| DCC1 | 0.72 | 0.79 |
| DCC2 | 0.85 | 0.88 |
| DCC3 | 0.91 | 0.92 |
| DCC4 | 0.86 | 0.87 |
| DCC5 | 0.78 | 0.81 |
| DCC6 | 0.80 | 0.84 |
| DCC7 | 0.74 | 0.77 |
| Green reporting capability (GRC): Cronbach's alpha: 0.89; AVE = 0.653; CR = 0.902 | | |
| GRC1 | 0.79 | 0.81 |
| GRC2 | 0.83 | 0.86 |
| GRC3 | 0.88 | 0.91 |
| GRC4 | 0.81 | 0.83 |
| Green sustainability disclosure (CSD): Cronbach's alpha: 0.88; AVE = 0.678; CR = 0.904 | | |
| CSD1 | 0.73 | 0.78 |
| CSD2 | 0.85 | 0.89 |
| CSD3 | 0.91 | 0.92 |
| CSD4 | 0.87 | 0.90 |

Abbreviations: AVE, average variance extracted; CFA, confirmatory factor analysis; CR, construct reliability; Cronbach's Alpha, reliability statistics; EFA, exploratory factor analysis.

tests were done in this study. Univariate normality was evaluated using the Shapiro Wilk test and multivariate normality was done using the Mardia test. The findings suggest that all the P-values have been below 0.05, and thus the data is more or less normally distributed, thus condoning the use of parametric tests like PLS-SEM to test the hypotheses.

### 3.6 | Common Method Bias

Since survey-based research is especially vulnerable to common method bias (CMB): it can either overstate or understate beta estimates, such that this study chose both procedural and statistical solutions to reduce the possibility of common method variance (CMV) so that the results can be robust. To begin with, the basic remedies being procedural amendments were introduced at the design level. They assured the respondents of complete anonymity and confidentiality to minimize the social desirability bias and evaluation apprehension. Measurement items were also carefully worded in order to prevent ambiguity, leading questions, and double-barreled statements. The order of questions was also randomly selected to minimize the respondent prediction of the relationship between constructs and prevent bias in the responses. The steps were used to minimize systematic covariance that was caused by measurement methodology and not by the measures themselves. Second, a number of statistical tests were carried out to determine the existence of CMV. The single-factor test conducted by Harman (Podsakoff et al. 2003) was conducted and the results of the test showed that none of the factors explained a significant proportion of the overall variance. CMV cannot be a very influential factor since there is no clear-cut result. Moreover, the measurement error was also managed by a multi-indicator method and Maximum Likelihood (ML) estimation in the ML framework of S.E.M. (Kline and Tamer 2016), boosting construct validity and minimising systematic measurement error. As a more rigorous test a Common Method Factor (CMF) was performed by comparing an unconstrained measurement model to a fully constrained model with a latent method factor. The difference between the two models was not significant ($\Delta\ \Delta\chi^2 = 0.152$, $\Delta df = 0.21$, $p = 0.78$) meaning that the inclusion of a method factor did not significantly increase the model fit. This can be seen to prove that there is no large shared variance that can be attributed to method bias in the dataset. Overall, such a mix of procedural safeguards and numerous statistical diagnostics is a good indication that CMB that can affect the estimated relationships is insignificant, thus making the reported beta coefficients and structural model results more credible and valid.

### 3.7 | Control Variables

Three control variables were considered to factor in the possibility of confounding effects: firm age, industry type, and firm size. The existing studies show that the nature of firms may affect strategy adaptation to technologies, sustainability, and disclosure performance results (Wiklund and Shepherd 2005). The size of the firms was determined in terms of the number of employees and the yearly turnover, which determines the available resources to adopt AIEA and GRC. Firm age incorporates organisational experience and maturity in its integration of the sustainability practices. Type of industry (manufacturing, service, retail) takes care of sector-specific regulatory, environmental and operational differences. It has been found that the



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**TABLE 3** | Descriptive statistics and correlation matrix.

| Constructs | Mean | SD | AIEA | DCC | GRC | CSD |
|---|---|---|---|---|---|---|
| AIEA | 4.06 | 0.94 | 1 | | | |
| DCC | 3.89 | 0.96 | 0.652* | 1 | | |
| GRC | 3.92 | 0.91 | 0.332* | 0.218* | 1 | |
| CSD | 4.03 | 0.93 | 0.584* | 0.498* | 0.369* | 1 |

*Correlation is significant at the 0.01 level.

**TABLE 4** | Discriminant validity.

| Constructs | Correlation | Squared correlation | AVE* | AVE > squared correlation |
|---|---|---|---|---|
| AIEA × DCC | 0.652 | 0.425 | 0.694 | Yes |
| AIEA × GRC | 0.332 | 0.110 | 0.677 | Yes |
| AIEA × CSD | 0.584 | 0.341 | 0.689 | Yes |
| DCC × GRC | 0.218 | 0.048 | 0.671 | Yes |
| DCC × CSD | 0.498 | 0.248 | 0.682 | Yes |
| GRC × CSD | 0.369 | 0.136 | 0.658 | Yes |

*Note:* AVE is the average of correlated constructs.

**TABLE 5** | Multicollinearity test.

| Constructs | Beta | *t* | Sig. | Tolerance | VIF |
|---|---|---|---|---|---|
| AIEA | 0.391*** | 6.512 | 0.000 | 0.505 | 1.980 |
| DCC | 0.275** | 3.874 | 0.000 | 0.540 | 1.852 |
| GRC | 0.203** | 3.123 | 0.002 | 0.670 | 1.493 |
| AIEA × DCC | 0.162** | 2.789 | 0.006 | 0.788 | 1.270 |
| AIEA × GRC | −0.041 | −1.055 | 0.293 | 0.870 | 1.149 |

*Note:* **$p < 0.001$, ***$p < 0.01$.

control variables did not have a strong influence on the model, and structural relationships between AIEA, DCC, GRC, and CSD were strong, which proves that the results of the study are stable and reliable.

### 3.8 | Multicollinearity

A linear regression analysis was carried out to investigate any possible multicollinearity between the independent variables and the interaction variables. The variability of the predictors evaluated through Tolerance and variance inflation factor (VIF) was used to determine the intercorrelations of the predictors. VIF gives the degree to which a regression coefficient variance is inflated as a result of collinearity with other predictors and should preferably be below 3 (Weston and Gore Jr 2006). On the other hand, tolerance is used to measure the percentage of the variance of a predictor that is not determined by other predictors, and it must be above 0.1. Table 5 shows the outcomes of the multicollinearity analysis of the constructs in this study. The values of VIF were 1.15–1.98 with tolerance values 0.505–0.870, which proves that multicollinearity is not an issue with the proposed model.

## 4 | Results

The standardised path coefficients are summarised in Table 6 and found to be the outcome of the structural model. Hypothesis 1 was that there should be a positive and direct effect of AIEA on CSD. This was proved with 0.332 at $p < 0.01$. The hypothesis of the positive and direct impact of GRC and DCC on CSD were proposed by Hypotheses 2 and 4 respectively. These two hypotheses were confirmed with 0.279 ($p < 0.001$) and 0.418 ($p = 0.001$) respectively. Moreover, the overall fit indices (2/df = 2.764; CFI =0.982; IFI =0.979; RMSEA =0.071; SRMR =0.061) also show the model fits very well. The indices are all within the reasonable range indicating that the data structure is well represented by a strong and reliable model suggested. Through the bootstrapping process, the mediating effect between AIEA ➜ GRC ➜ CSD was looked at indirectly. These results indicate that AIEA has a significant impact on the dependent variable CSD (0.332, $p < 0.01$) and on the mediator variable GRC (0.287, $p < 0.001$). Moreover, GRC has a significant influence on CSD (= 0.279, $p$ =. 001). This proves that to some extent GRC mediates between AIEA and CSD. AIEA ➜ GRC ➜ CSD was also found to be significant (0.092, $p < 0.001$), which supports Hypothesis 3. The scale of the robustness of the indirect, direct and total effects was confirmed by the bootstrapping technique with 5000 resamples and a 95% confidence interval (CI) which is according to the recommendations made by Hayes and Scharkow (2013). Therefore, mediation effect was statistically confirmed.

### 4.1 | Moderation Effect

To validate Hypotheses 5 and 6, a two-stage interaction modelling was conducted as suggested by Kenny and Judd (1984). In the initial process, the variables in the study were standardised in IBM SPSS generating standardised values of each



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**TABLE 6** | Summary of hypothesis testing.

| Hypothesis | Standardized coefficient | | Remark | $R^2$ |
|---|---|---|---|---|
| | Direct effects | Indirect effects | | |
| Independent effects | | | | |
| Hypothesis 1: AIEA > CSD | 0.332** | — | Supported | |
| Hypothesis 2: GRC > CSD | 0.279*** | — | Supported | |
| Hypothesis 4: DCC > CSD | 0.418*** | — | Supported | |
| Mediation Effects | | | | |
| Hypothesis 3: AIEA > GRC > CSD | — | 0.092*** | Supported (partial mediation) | |
| Moderating Effects | | | | |
| Hypothesis 5: DCC × AIEA > CSD | 0.167** | — | Supported | |
| Hypothesis 6: DCC × GRC > CSD | 0.124* | — | Supported | |
| GRC | | | | 0.081 |
| CSD | | | | 0.462 |

*Note:* *Significant at < 0.001 level; **Significant at < 0.01 level; ***Significant at < 0.05 level.
Abbreviation: $R^2$, squared multiple correlations.

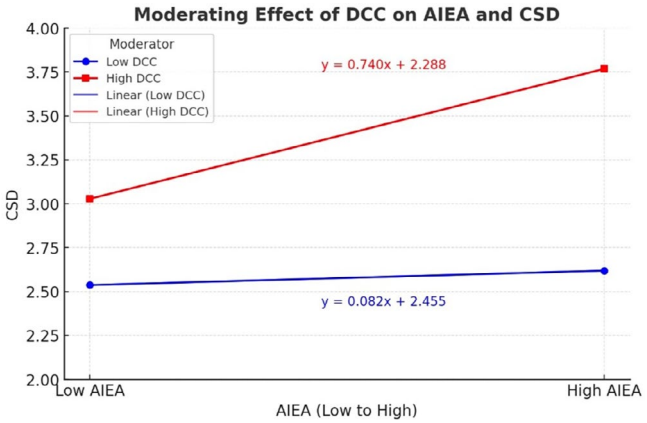



**FIGURE 2** | Interaction of DCC and AIEA.

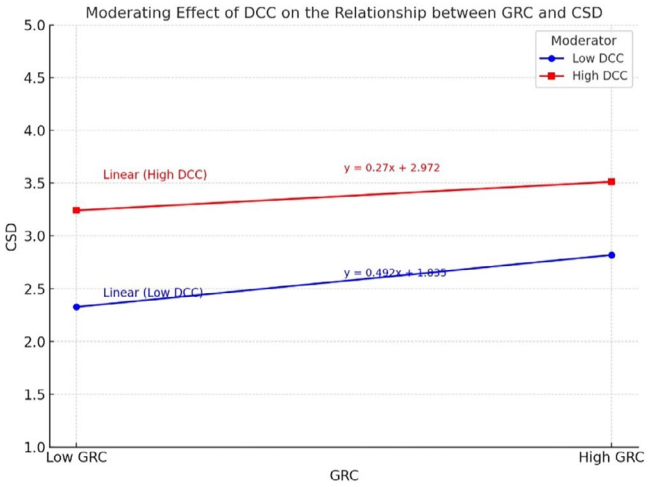



**FIGURE 3** | Interaction of DCC and GRC.

construct; that is, AIEA, GRC, and DCC. The second stage involved the multiplication of the standardised variables to generate two product terms namely DCC × AIEA and DCC × GRC. These terms of interaction were inputted to the model and tested together with the main effects. The findings indicate that the interaction term DCC × AIEA and CSD have positive and significant correlation with standardised coefficient of 0.167 and $p = 0.01$. It establishes that DCC enhances positive influence of AIEA on CSD hence Hypothesis 5. In the same manner, DCC × GRC was reported to have a strong positive correlation with CSD (= 0.124, $p = 0.05$). It indicates that DCC increases the influence of GRC on CSD, Hypothesis 6 supported. Figure 4 shows the structural model in graphic form and Table 6 gives the summary of the results of the hypothesis tested. Figures 2 and 3 were used to plot the interaction effects to visually interpret the patterns of moderation. Figure 2 shows that the relationship between DCC and AIEA is that as organisations express high rates of DCC, the relations between AIEA and CSD increase significantly. On the other hand, in low concentration of DCC, this effect is not strong. In this way, DCC enhances the positive effect of AIEA on CSD, and there is a good indication of Hypothesis 5. In Figure 3, DCC and GRC interaction depicts that the higher the level of DCC in organisations, the greater the effect that GRC has on CSD. This removes any doubt of the fact that DCC, in addition to enhancing the effectiveness, multiplies the efficacy of GRC to enhance sustainability disclosures, and thus, confirms Hypothesis 6. The results show that the structural model (Figure 4) has a great fit and good predictive validity. There were found to be significant all direct and indirect relationships among the AIEA and GRC, DCC, and CSD, thus, proving the hypothesised relationships. In particular, Hypotheses 1, 2 and 4 were tested and maintained, which proved the positive impact of AIEA, GRC, and DCC on CSD. On the same note, Hypothesis 3 was accepted confirming the partial mediating GRC. Moreover, both Hypotheses 5 and 6 were accepted, and the positive moderating effect of DCC on the effects of AIEA and GRC on CSD was proven. To sum up, the results indicate



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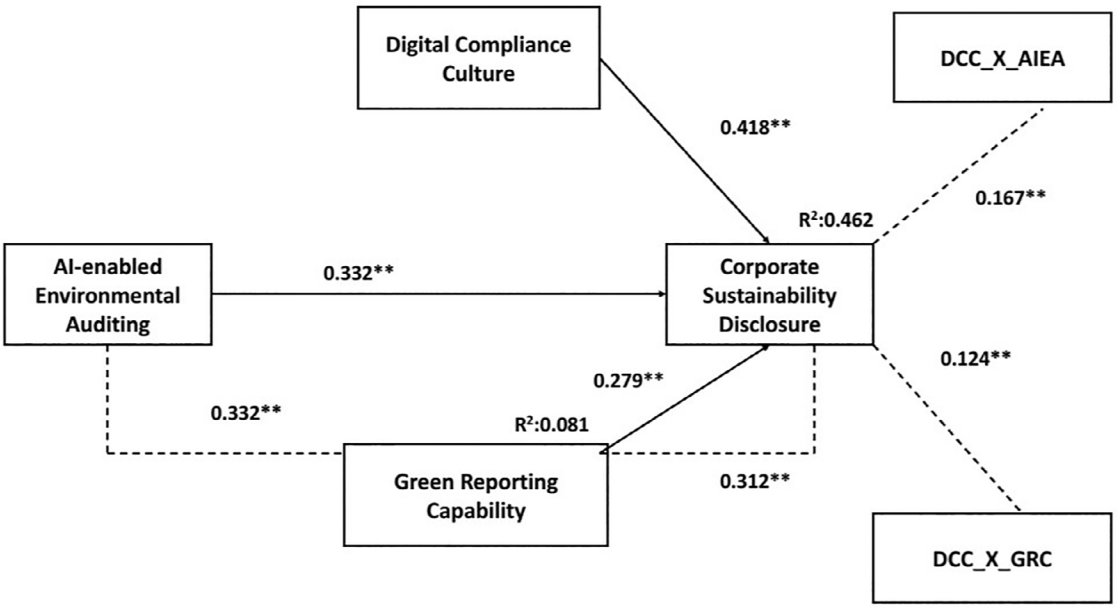



**FIGURE 4** | Structural model. $^{**}p < 0.01$, $^{***}p < 0.00$. $R^2$, squared multiple correlations.

that AIEA and green reporting systems, integrated into a DCC can enhance CSD practices significantly.

The usefulness of the structural relationships was established through the calculations of effect size (f2). The impact of GRC on CSD was the highest as compared to AIEA, although the direct and moderating impact of DCC was significant. These findings suggest that the relationships observed are statistically as well as practically important. The $Q^2$ values were all positive, and predictive power was found in-sample. Out-of-sample predictive performance based on PLS-Predict indicated that most of the RMSE values of the PLS model were lower than the values of a linear benchmark model showing moderate-strong predictability ability of the model in predicting CSD.

Values representing the effect sizes (f2) and the PLS-Predict RMSE are not presented to prevent crowding out of the Results section. The impact and predictive accuracy are well reported and explained in the text, and complete numerical values may be given in Supporting Information where appropriate. This is so as to provide clarity, conciseness, and readability and does not affect the impact and completeness of the findings.

## 5 | Discussion

This paper provides a theoretically and empirically validated examination of the interaction between AIEA, GRC, and DCC in establishing CSD among South Asian SMEs. This combination of RBT, Legitimacy Theory, and DCT makes the findings a complete description of how the digital environmental governance processes can be transformed into plausible and transparent sustainability reporting.

### 5.1 | H1: Aiea → CSD

The findings confirm that there is a significant and positive correlation between AIEA and CSD. This outcome supports the strategic role of AIEA as a valuable, rare, and inimitable resource, which is aligned with RBT (Fosso-Wamba et al. 2024; Khan et al. 2024). AI-based environmental auditing can improve the real-time tracking of environmental performance, enable the tracking of emissions, and align environmental indicators with sustainability indicators (Tomar 2025). AI systems help decrease reporting errors, increase data reliability by minimizing the manual input (Lu and Li 2025; Tarawiru et al. 2025). In terms of the Legitimacy Theory, the introduction of AI is evidence of technological proficiency, transparency, and proactive leadership, which will reinforce stakeholder trust and alignment with society (de Almeida and Oliveira 2024; Kant et al. 2025; Quan 2025). The apparent inclusion of AI in the sustainability reporting process improves the perception of responsibility and accountability in a South Asian environment where sustainability reporting is usually voluntary. Though the focus on the use of AI in multinational corporations has been dominant in the previous researches (Aldhi et al. 2025), this study indicates that AI can be used as a strategic equalizer by resource-constrained SMEs. This increases the empirical coverage of AI and sustainability literature and corrects the assumption that only big companies can have sophisticated environmental auditing systems (Zhou et al. 2022) However, as positive as the direct effect of AIEA on CSD, the results indicate that technological resources



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do not always optimize the quality of disclosures, which implies the use of internal reporting competencies.

### 5.2 | H2: GRC → CSD

The findings suggest that the impact of GRC on CSD is significant and positive, which supports the role of internal reporting competencies as long-term sources of competitive advantage (Gazi et al. 2024; Neiroukh and Çağlar 2025; Quan 2025). GRC can help companies to create credible, similar, and practical sustainability reports, which enhance stakeholder legitimacy (Lai 2025). Traditional beliefs imply that SMEs have a disadvantage of inadequate reporting systems as a result of limited resources (Shen et al. 2025). This has however been challenged by the current results that indicate that SMEs that have formalized GRC systems are able to deliver an equivalent and even better disclosure performance than larger firms (Damnjanović et al. 2025). This finding disrupts firm-size determinism in sustainability reporting literature and introduces a more nuanced understanding of SME capacity for transparency. Importantly, the positive role of GRC complements the findings of Hypothesis 1. While AIEA provides technologically advanced environmental data, GRC structures and standardizes this data into coherent, stakeholder-oriented sustainability disclosures (Mervelskemper and Streit 2017). This interplay between resource and capability aligns closely with RBT and sets the stage for examining the mediating mechanism proposed in Hypothesis 3.

### 5.3 | H3: AIEA → GRC → CSD (Mediating Role of GRC)

The intermediary role of GRC substantiates the fact that AIEA is ineffective on its own to maximize the effect of CSD. According to DCT, technological resources can be strategically valuable only in the cases when firms have these dynamic capabilities to interpret, integrate, and institutionalise (Tomar 2025; Fosso-Wamba et al. 2024). By using GRC, SMEs engage AI-generated knowledge into their official reporting structures, cross-functional environmental information, and transform disclosure practice according to changing regulatory and stakeholder expectations (Kant et al. 2025; Lu and Li 2025). It is a dynamic process that indicates the transformation of sensing (AI-based data acquisition) to seizing and reconfiguring (reporting routines and disclosure adaptation). The need to fill this critical gap in the literature is attributable to the fact that previous studies have mostly conceptualised AI and reporting capabilities independently without testing the mediating effect of GRC empirically (Neiroukh and Çağlar 2025). The study contributes to the theoretical contribution of RBT and DCT in sustainability governance studies by showcasing the fact that GRC is the mechanism that AI can help to improve the quality of disclosures.

### 5.4 | H4: DCC → CSD

The direct impact of DCC on CSD justifies the relevance of ethical, responsible, and technological compliance cultures in the formation of sustainability disclosure quality. Considering the Legitimacy Theory perspective, a good DCC is an indication of high data governance and environmental stewardship, which increases the level of credibility even in a relatively voluntary reporting context like in South Asia (Lai 2025). Notably, the results do not confirm that compliance cultures hamper innovation. Rather, DCC is an enabling tool, which facilitates transparency, accountability, and quality disclosures (Lu et al. 2024). This re-packages compliance not as a bureaucratic dogmatism but as a strategic cultural infrastructure, which promotes believable sustainability communication.

### 5.5 | H5: Moderating Role of DCC on AIEA → CSD

The relationship between AIEA and CSD is enhanced by the moderating effect of DCC. By incorporating the digital standards of compliance into the organisational culture, AI-generated insights are implemented systematically, processed ethically, and regularly incorporated in the reporting practices (Kant et al. 2025). Without DCC, AI systems can generate data that is not fully used or reported irregularly. Nonetheless, a high compliance culture is a guarantee of disciplined execution, accountability across functions, and transparency (Long et al. 2023). This observation supports DCT as it proves that cultural preparedness makes technological resources become more strategic.

### 5.6 | H6: Moderating Role of DCC on GRC → CSD

In the same vein, DCC augments the good correlation between GRC and CSD. DCC enhances the cross-functional alignment, internal responsibility, and compliance with reporting standards to convert reporting capabilities into timely, plausible, and credible sustainability disclosures (Li and Jin 2024). This exchange also contributes to the postulations of DCT that capabilities are best produced in an enabling organisational context. Cultural preparedness is therefore a booster, which increases the efficacy of the technological systems and reporting structures (Elamer and Boulhaga 2024). Overall, the synthesis of RBT, Legitimacy Theory, and DCT leads to a whole model of digital sustainability governance in SMEs. AIEA is the strategic technological enabler, GRC is the dynamic organisational enabler, and DCC is the strategic cultural enabler. All these factors contribute to the strengthening of CSD by direct, mediated, and moderated mechanisms. The findings complement the current body of literature by showing that sustainability transparency is not possible by the means of technology only. Rather, organisational capabilities should be instilled with technological systems and supported by the culture of digital compliance (Bin-Feng et al. 2024). The study of these interdependencies empirically leads to a new level of theoretical discussion and offers a solid rationale regarding the possibility of attaining credible and transparent sustainability disclosure by the SMEs in the emerging economies.

### 5.7 | Theoretical Implications

To describe CSD, the paper identifies several theoretical implications with the help of the constructs AIEA, GRC, and DCC. The combination of the RBT, Legitimacy Theory, and DCT allows



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developing more theoretical knowledge about the technological, organisational, and cultural processes underlying sustainability reporting.

To begin with, the study also builds on the RBT by giving it empirical evidence that AIEA is a valuable, rare, and inimitable technological resource that can improve sustainability disclosure even in SMEs. Even though the previous conceptual schemes were inclined to the view of AI as a technical resource, which can be deployed by large companies only, this study demonstrates that AI-based auditing systems should be defined as strategic resources that can generate timely, credible, and decision-relevant environmental intelligence. These technological capabilities are more strategic in a resource-constrained situation, when they balance the lack of human and financial resource and allow SMEs to compete in terms of transparency and accountability. GRC as a mediating factor is an important contribution to the theory. The results show that AI-generated information in itself is not enough; it should be incorporated into organisational practices that codify, harmonise, and commercialise environmental information into organized disclosures. GRC thus is an organisational level dynamic capability through which the technological inputs are converted into strategic reporting outputs. This is quite consistent with the DCT which assumes that sustainable competitive advantage is not only the result of resource endowment, but rather, of the capacity of the firm to combine, reorganize, and mobilize resources in accordance with the needs of the environment. Within this context, Ai-Assisted Environmental Auditing fulfils the sensing role, GRC is seizing and reconfiguring, and the disclosure results are institutionalised operationalization. Through the process approach of understanding this sequential mechanism, the study provides a process explanation of how the technological potential is converted into the sustainability transparency. Notably, the paper provides significant elaboration and reinforcement of the usage of the Legitimacy Theory, specifically, the author builds on the original conceptualization of the idea of corporate sustainability reporting that Camilleri (2018) reflects on, noting that the inherent characteristic of the discussed concept is the fact that it is a mechanism with which a company retains, restores, and reinforces social legitimacy under the conditions of changing stakeholder expectations. Following Camilleri (2018), legitimacy is not an absolute concept, it is created socially, it is a negotiable construct, and depends on the transparent dialogue about responsible behaviour. It is based on this perspective, that the current research contributes to the theory of Legitimacy in three significant aspects. To start with, it shows that the legitimate technology adoption does not necessarily create the legitimacy. The very fact of the implementation of AI systems is not a guarantee of better stakeholder trust. Rather, legitimacy is achieved as a synergistic relationship between technological (AI-enabled auditing), organisational (GRC) and institutionalized cultural norms (DCC). This goes beyond the symbolic legitimacy, where companies implement visible technologies to participate in impression management, to substantive legitimacy, which is based on believable, data based and structured disclosures. Secondly, DCC adds to the Legitimacy Theory by introducing the idea of organisational culture as a strategic legitimacy-enabling tool. Camilleri (2018) asserts that the stakeholders are becoming more demanding to have true incorporation of sustainability in the fundamental activities in an organisation. DCC operationalizes this expectation through the reflex of helping to instill ethical norms, accountability standards, cross-functional coordination, and routines that are technologically made into the daily operations. Being a moderating variable, it enhances the correlation between AI auditing, reporting capability and disclosure quality and makes sure that the sustainability communication is not a facade of a superficial reporting, but a reflection of organisational commitment. Therefore, the institutionalisation of legitimacy is achieved internally and then transferred to the outside world. Thirdly, the paper reformulates CSD as a legitimacy-generating process and not a reporting achievement. Continuous sensing (AI data acquisition), seizing (capability development), and institutionalizing (culture embedding) are the ways to attain legitimacy. This is a multi-level and hierarchical framework that extends the Legitimacy Theory into a process-oriented and hierarchical framework that unites it with RBT and DCT to construct an effective explanatory framework. The three-way structure is in a position to measure both direct and mediated-moderated relationships, providing a more detailed perspective of the way, in which the legitimacy is created by technological modernization and change of the organisation. The combination of culture and resources and capabilities helps the study to no longer be linear and single theory and to suggest a configurational legitimate architecture where AI-Based Environmental Auditing as part of technological sensing, GRC as part of seizing and reconfiguring, and DCC as part of institutionalising and normative alignment.

Lastly, the study enhances the empirical basis of sustainability digitalization literature because it quantitatively tests mediating and moderating relationships by applying structural equation modelling (SEM). This empirical confirmation goes further than the conceptual argumentation to statistically supported interdependences between technology, capability, and culture. By doing so, the study offers new empirical data to a relatively unexplored territory where the integration of AI, the ability to report sustainably, and the organisational culture has been considered as a conceptual phenomenon. Altogether, in terms of explaining Legitimacy Theory in accordance with Camilleri (2018) and integrating it into the framework of a dynamic, resource-capability-cultural viewpoint, the study is a solid, powerful, and valuable contribution to the study of sustainability management and corporate disclosure.

### 5.8 | Practical Implications

This research has practical and strategic suggestions that managers, policymakers and practitioners may employ to promote CSD, particularly in resource limited SMEs in developing economies. The study directs its attention to AIEA, GRC, and DCC as the points of intervention and enhances the transparency, trust, and reliability of sustainability reporting on a technological, organisational, and cultural level. As a manager, the findings present the strategic importance of investing in AIEA. AI may be used to automate data collection, monitoring, and verification of environmental indicators, minimize human errors, enhance traceability, and make the data record audit-ready. Managers must hence take into consideration the adoption of AI in organisational operations as a means of guaranteeing a uniform, precise, and timely reporting of the environment. Nevertheless,



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the study also points out that AI is not enough. This requires effective deployment, which is facilitated by building the GRC that helps organisations to codify, standardise, and interpret AI-generated insights into actionable and credible disclosures of sustainability. In the case of SMEs, this can be in the form of training employees, computerized reporting systems, and internal controls, and also by involving external expertise to develop reporting capability in stages. It is also of great essence to have a strong DCC as a strategic enabler. This kind of culture instils responsibility, ethical digital citizenship, transparency, and digital literacy in the organisational culture, which supports the operationalisation of AI systems and reporting functions. Some of the practical solutions are formal policies, integration of compliance measures in performance appraisal, and regular training on responsible digital practices. Combined, AI, GRC, and DCC ensure that sustainability reporting is credible and consistent with the expectations of the stakeholders even where SMEs have limited resources. To the policymakers and regulators, the results imply specific institutional responses such as providing SMEs with the necessary institutional support in their use of AI-based auditing tools. These can be subsidies, templates of standardised digital reporting, capacity-building activities, and public-private partnerships. Sustainability practices can also be promoted by encouraging ethical, open, and digitally enabled practices, which can be enhanced through certification schemes and industry-driven movements to improve governance internally and beyond national borders. Overall, the research provides valuable advice to sustainability consultants, technology developers, and industrial associations. Consultants are able to guide the SMEs in matching technology adoption to internal capacity and cultural enablers. Technology developers have an opportunity to work on the development of user-friendly AI auditing solutions that are compatible with the current reporting systems, and thus specific to the SMEs that are less digitally mature. Knowledge-sharing networks can be promoted by industry associations to share the best practice in AI adoption, reporting capabilities building, and digital frameworks of compliance. In general, the analysis shows that quality sustainability disclosure is the result of the synergetic technological, organisational, and cultural processes. By operationalising AI-generated insights with GRC and formalising them with an effective DCC, SMEs are able to address resource shortages, minimize the chances of reporting mistakes or greenwashing, and enhance stakeholder confidence.

## 6 | Conclusion

This research contributes to the existing body of knowledge on the relationship between AIEA, GRC, and DCC and the nature of their role in shaping CSD by studying how these three notions of digital sustainability governance in SMEs interrelate. Findings indicate that although AIEA helps enhance the accuracy, monitoring, and transparency, its success relies on GRC that would convert AI-based insights into actionable, structured, and credible disclosures. Furthermore, DCC performs as a cultural enforcer, enhancing both technological and reporting systems through incorporation of accountability, ethical behaviour, as well as transparency into organisational practices. A combination of these results shows that realistic sustainability reporting is the result of a combination of technology, internal competence, and cultural readiness and provides SMEs with a feasible framework for improving the quality of reporting, stakeholder trust, and legitimacy.

In terms of the scope, the results are contextually based on the South Asian SMEs, wherein regulatory implementations, digital maturity, and institutions are not comparable to developed economies. Although the mentioned mechanisms (AIEA, GRC, and DCC) might be applicable in other SME-related settings, one should be wary of the generalizability of the findings to the other similar emerging markets. Future research may be used to evaluate the applicability of the model to more technologically advanced, highly controlled, or culturally different areas to determine its applicability in a wider setting and further develop theoretical knowledge about AI-enabled sustainability practices in different institutional environments. Overall, the study adds to the existing literature covering ESG and AI disclosure by incorporating technology, organisational competence, and cultural enablers as one validated model, which offers both theoretical and practical importance to scholarship, managers, and policymakers.

### 6.1 | Limitations and Future Research Directions

Although this research contributes to the realization of the functions of AIEA, GRC, and DCC in the creation of CSD extensively, one must admit a number of limitations. First, the study is only conducted among the SMEs of the developing economies where the regulatory environment, institutional background, and digital infrastructure are significantly different as compared to the developed environment. This means the generalizability of the results to more controlled or more technologically developed settings might be reduced. Future studies might measure whether such relationships are true in larger companies, multinational corporations, or SMEs in the regions with more strict compliance frameworks and with a higher level of digital maturity.

Second, the study is cross-sectional, which means it involves organisational practices at one time only. Although this gives important insights on the present condition of the current AI-enabled sustainability practices, they do not factor in their development and long-term impacts. It is proposed to conduct longitudinal studies to investigate the evolution, stabilization, and learning about digital sustainability practices over time as it would have a dynamic view on the interaction between AI-Enabled Enviromental Auditing, reporting capabilities, and culture to drive CSD. Third, the study mainly dwells on the internal aspects of the organisation itself and fails to integrate the external forces of the supply chain partners, regulatory authorities, industry contacts, or pressures of stakeholders. A multi-level or ecosystem perspective may be taken up by future studies in order to capture these interactions and explain the joint impact of organisational and external forces on the disclosure of sustainability. Fourth, the individual-level variables, such as the level of digital skills, attitudes of the employees towards technology, and their ethical orientations were not taken into account. It might be beneficial to combine behavioural or psychological theories that would give a more in-depth understanding of the role of human factors in the adoption, operationalization, and effectiveness of AI-driven sustainability practices. Lastly, although the paper



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concentrates on AIEA as a disruptive technology, new digital technologies like blockchain, Internet of Things (IoT), and predictive analytics can also make remarkable contributions to the sustainability reporting. The combination of these technologies and their possible substitution of AI-Enabled Environmental Auditing as a more comprehensive view of digital sustainability governance and the relationship between technology, capabilities, and culture are subjects of future research. The limitations can be overcome by considering the limitations in future research; the findings can promote theoretical strength and practical implications, providing a more comprehensive depiction of the interplay of technology, organisational capabilities, cultural enablers, and external drivers leading to high-quality corporate disclosure of sustainability.


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### Ethics Statement

Informed Consent: Informed consent was obtained from all subjects involved in this study. Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki, and approved by the Ethics Committee of Shaheed Benazir Bhutto. University-(ASRAB/SBBU/-1RB00012248) and date of approval: September 12, 2025.

### Data Availability Statement

Complete data is reported in the manuscript itself.

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